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ORIGINAL RESEARCH

Data-Based Decision Making to Improve Organization, Time Management, and Planning: Screening and Progress Monitoring Tools

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Abstract

Improvements in organization, time management, and planning (OTMP) skills have downstream effects on student academic performance. Brief, practical assessment tools are needed to identify students needing OTMP intervention and monitor response to intervention. The purpose of this study was to develop and validate brief rating scales of OTMP skills, derived from the Children's Organizational Skills Scale parent and teacher versions (COSS-P, COSS-T), for screening and progress monitoring. Three samples were used: (1) general and clinical sample (n for caregivers = 1155; n for teachers = 1139; Abikoff & Gallagher, 2009); (2) sample from the randomized controlled trial (RCT) evaluating Organizational Skills Training-Tier 2 (OST-T2; n = 185; Nissley-Tsiopinis et al., 2024); (3) sample from the RCT evaluating Homework, Organization, and Planning Skills (HOPS; n = 163; Langberg et al., 2018). Based on indices of discrimination and item effect sizes, we identified 10 items from the COSS-P/T representing multiple components of OTMP interventions, in addition to one item assessing OTMP interference with performance. Confirmatory factor analyses demonstrated strong fit to a bifactor model. Only the general factor demonstrated acceptable reliability. The total score demonstrated convergent validity with measures of school and executive functioning and was excellent in differentiating children with and without substantial, impairing OTMP deficits (Area Under Curve [AUC] = .947 for COSS-P, .982 for COSS-T). Treatment sensitivity for the screeners and three, five-item formative behavior assessment measures, computed as effect sizes (Cohen's d) based on response to OST-T2 and HOPS, all exceeded −1.20. The brief tools developed in this study have substantial utility for screening and progress monitoring.

Keywords Organizational skills · Screening · Progress monitoring · Formative assessment · Data-based decision making

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Introduction

Executive functioning (EF) refers to a set of self-regulatory processes that enable individuals to plan and execute goal-directed behaviors. EF in childhood is associated with critical aspects of health and development, including behavioral health, social functioning, learning, and academic achievement (Barkley, 2020; Best et al., 2011; Nigg, 2016). Efforts to improve basic cognitive processes associated with executive functioning (e.g., working memory) have had limited impact on daily functioning (Chacko et al., 2014), but targeting practical, specific behavioral manifestations of EF, in particular organization, time management, and planning (OTMP) skills, has led to considerable success (Bikic et al., 2017). The accurate assessment of OTMP skills is important because OTMP skill level is predictive of later academic achievement (Langberg et al., 2011, 2013) and identifying children who could benefit from evidence-based interventions can alter their trajectory. Research supports the effectiveness of OTMP interventions to improve targeted, proximal outcomes (i.e., OTMP skills) as well as more distal outcomes (i.e., homework and academic performance) at the upper elementary, middle school, and high school levels (DuPaul et al., 2021; Evans et al., 2016; Nissley-Tsiopinis et al., 2024). Further, there is evidence that improvements in OTMP skills mediate the effect of OTMP interventions on academic performance (Evans et al., 2021; Pfiffner et al., 2013; Power et al., in press). In addition, OTMP interventions may reduce family conflict and improve students' attitudes toward school and teachers (Abikoff et al., 2013).

The contribution of OTMP skills to student academic performance, and in some research family relationships and attitudes toward school, underscores the importance of assessment tools to measure these skills. Measures are needed to identify students with OTMP deficits in need of intervention, formulate tailored intervention plans based on student strengths and weaknesses, monitor student progress in acquiring OTMP skills, and evaluate the outcomes of OTMP interventions. Numerous laboratory and behavioral assessment measures have been developed to assess EF and some of its behavioral manifestations (see Ryan et al., 2024 for a review). The Children's Organizational Skills Scale (parent and teacher versions; COSS-P, COSS-T; Abikoff & Gallagher, 2009) is unique in that: (a) it assesses specific behavioral manifestations of EF that have been shown to be predictive of academic achievement (Langberg et al., 2011), and (b) examines a broad range of skills manifested in real-world conditions at home and school. The COSS, developed for students in grades 2–8, assesses behaviors that are clearly observable and cut across the broad range of OTMP domains, including keeping track of assignments, organizing materials, managing time, and planning projects (Gallagher et al., 2014). Current versions of the COSS, derived from the original version (Abikoff & Gallagher, 2009) include 26 and 28 items for the COSS-P and COSS-T, respectively (Ryan et al., 2024). Although these scales have been shown to be useful for outcome assessment (Abikoff et al., 2013; Langberg et al., 2018; Nissley-Tsiopinis et al., 2024), they may not be efficient for use in screening a group of students or monitoring their response to intervention.

The overall goal of this study was to develop and validate brief versions of the COSS for the purposes of screening and formative assessment. Consistent with the unified theory of validity, the emphasis was on demonstrating that scores generated by the measures are meaningful and feasible for the purpose for which they are used, as opposed to merely describing psychometric properties of the measure (Messick, 1989). Measures are developed for a particular purpose; the validity argument is the statement of how to demonstrate that measures are effective for this purpose, and the validity evaluation is the assessment of whether the measure is actually effective in providing meaningful information (Kane, 2013). The following sections describe the validity argument that guided the approach used in this study to develop and validate brief, meaningful measures for the purpose of screening and formative assessment.

Development of Screening Measures

Numerous factors can contribute to the selection of items from established teacher- or parent-rated measures to include in a screening instrument. First, it is advisable to include items with relatively high pattern coefficients or factor loadings on salient factors identified in factor analyses of established measures (Power et al., 2013), which indicates that items are aligned with important constructs assessed by the established measure (Floyd & Widamon, 1995).

Second, given that a major purpose of screening is to identify individuals meeting criteria for having a problem, it is critical to demonstrate that items discriminate adequately between subgroups who likely have the problem from those who likely do not have the problem (Ebel, 1979; Youngstrom, 2014). One useful method for evaluating how well an item differentiates individuals with varying levels of a characteristic is by examining frequency distributions for each item (i.e., ensuring a broad range of response options are used by informants). Another useful method is computing each item's index of discrimination, which indicates the difference in proportion of respondents endorsing an item between the subgroups scoring in the highest and lowest quartile on the scale (Ebel, 1979).

Third, given that screening measures are often used to identify potential targets of intervention for students, it may be useful to select items aligned with key components of

corresponding evidence-based interventions (Daniels et al., 2014). For example, key components of OTMP interventions include keeping track of assignments, managing materials, managing time, and planning (Gallagher et al., 2014). As such, it is advisable to select one or more items that map on to each of these core components of intervention. A related consideration pertains to the inclusion of items with a reasonable degree of responsiveness to the effects of intervention (i.e., treatment sensitivity). Treatment sensitivity is commonly evaluated by examining within-group change from the baseline to the intervention period (Vannest et al., 2010 An alternative, potentially more stringent test, is to examine between-group change during the intervention period between a group who receives the intervention and a control group who receives treatment-as-usual or an alternative intervention (Matta et al., 2020).

Additional considerations for selecting items include the extent to which the item makes a unique contribution to the measurement of the construct(s) being examined. Examining the pattern of item intercorrelations within informant can be useful for this purpose. Items assessing the same broad construct (e.g., executive functioning) would be expected to correlate with one another to a moderate degree. When an item exhibits a high correlation with another item (>.80), it is reasonable to question whether both items should be included in the screening measure because of the extent of redundancy. In contrast, when a screening measure is being developed for multiple informants, there are advantages to including similar items across informants to facilitate an examination of student functioning in different contexts (De Los Reyes et al., 2022), including OTMP behaviors at home and in school.

Development of Progress Monitoring Measures

Progress monitoring is critical for data-based decision making in that it provides valuable information about whether an intervention is working, which targeted behaviors appear to be changing, and which behaviors may not be changing. This information is highly useful in determining whether changes to the intervention plan are needed. For research in schools, there is a long, successful history of monitoring the progress of instructional interventions by applying curriculum-based measurement methods. Since about 2005, there has been rapid growth in progress monitoring of social and behavioral functioning as well as academic enablers (e.g., academic engagement, study skills, organization; Chafouleas et al., 2007; Volpe et al., 2023). To serve this purpose, researchers have developed and validated Direct Behavior Ratings (DBRs), which assess a behavior that has been operationally defined and measured after a relatively short period of time (e.g., daily rating at end of school day; Chafouleas et al., 2007). Most of the research on DBRs has focused on the use of well-defined single-item scales (DBR-SIS). More recently, research has focused on the development and validation of DBR multi-item scales (DBR-MIS; Volpe & Briesch, 2012), which typically include three to five items. Items to be included on DBR-MIS may consist of a standard set of items, a set of items tailored to specific students based on their functioning at baseline, or a hybrid of these two approaches (Volpe & Briesch, 2015). Both approaches (DBR-SIS, DBR-MIS) have demonstrated their value and psychometric soundness in the formative assessment of disruptive behavior, internalizing functioning, interpersonal competence, and academic enablers (see Volpe et al., 2023). Although DBRs are designed to be completed daily or even more frequently, these methods have been successfully adapted for use on a weekly basis or less often (referred to as formative behavior rating measures [FBRM]; Volpe et al., 2023). Volpe et al., (2010) proposed several methods of extracting subsets of items from extant screening measures with favorable psychometric characteristics for progress monitoring, highlighting the need for behavioral assessment of both performance and general objectives to measure changes in specific target behaviors and broader constructs in response to intervention, respectively. Many of the factors to be considered in selecting items for screening instruments also apply in developing progress monitoring tools. In particular, it is important for items to align with the targets of corresponding evidence-based interventions. Another critical factor is that items should have a reasonable degree of treatment sensitivity, demonstrated through within-group or between-group analytic methods (Matta et al., 2020; Volpe et al., 2023). In addition, factors such as the observability, malleability, and social validity of assessment targets, as determined by practitioner and scientific advisory panels, may be useful in selecting items for formative assessment (Volpe et al., 2020).

The purpose of this study was to develop measures for screening and progress monitoring related to the assessment of OTMP skills to facilitate data-based decision making. The study was conducted in two parts. The first part (i.e., Study 1) focused on *measure development* and involved the selection of items for screening and formative assessment (progress monitoring), and the second part (i.e., Study 2) focused on *measure validation* of the screening and formative assessment measures. Candidate items for the screening and formative assessment measures were selected from the subsample of items from the original versions of the COSS that included 26 items of the COSS-P and 28 items of the COSS-T, which were identified and confirmed through a series of exploratory and confirmatory factor analyses (Abikoff & Gallagher, 2009; Molitor et al., 2017; Ryan et al., 2024).

Study 1: Measure Development

Method

Study 1 was guided by three aims. Aim 1 focused on examining how well each item of the COSS-P and COSS-T discriminated students likely to have elevated OTMP deficits from those who did not. Aim 2 examined the treatment sensitivity of each of the COSS-P and COSS-T items, and Aim 3 identified the alignment of each COSS item with the components of evidence-based OTMP interventions.

Participants

The aims for Study 1 were addressed using three separate data sets. The data set used to evaluate item discrimination (Aim 1) was derived from the general and clinical samples used to standardize and evaluate the psychometric properties of the COSS (Abikoff & Gallagher, 2009), which was collected between 2004 and 2007. In the general and clinical samples, about 300 children were rated by both caregivers and teachers; the remaining students were rated by only one informant. The total sample included 1155 children rated by caregivers on the COSS-P and 1139 students rated by teachers on the COSS-T. The general/non-clinical subsamples consisted of 762 (66%) caregiver-rated students and 877 (77%) teacher-rated students; the remaining children were in the clinic-derived subsamples, which consisted of a relatively high percentage of children with ADHD (87% for caregiver-rated children; 65% for teacher-rated children). The non-clinical subsample was recruited to resemble the racial/ethnic composition of the United States in 2000. For the total samples, children ranged in age from 8 to 13; the mean age of the children in the caregiver- and teacher-rated samples was 10.5 ($SD = 1.7$) and 10.8 ($SD = 1.7$), respectively. Assigned sex of the children was 43.9% female and 48.1% female in the caregiver- and teacher-rated samples, respectively. As reported by Ryan and colleagues (2024), in the caregiver-rated sample, 10.4% of the children were Black, 6.6% Asian, 12.6% Latiné/Hispanic, 0.5% American Indian/Alaskan Native, 4.8% Multi-racial, 64.2% White, and 0.9% were not reported or indicated as Other. In the teacher-rated sample 14.7% were Black, 9.0% were Asian, 12.6% were Latiné/Hispanic, 0.4% were American Indian/Alaskan Native, 4.6% were Multi-racial, 57.2% were White, and 1.7% were not reported or Other. In the caregiver-rated total sample (including the general and clinical subsamples) 29.3% of the children were reported to have ADHD, and in the teacher-rated sample, 16.1% reportedly had ADHD. Additional information about the clinical and non-clinical subsamples, in combination and separately, was described in prior research (Abikoff & Gallagher, 2009; Ryan et al., 2024).

Information used to examine the treatment sensitivity of COSS items (Aim 2) was derived from two data sets. The first included students in grades 3–5 participating in a cluster-randomized trial examining the effectiveness of the Organizational Skills Training – Tier 2 program (OST-T2) conducted in 22 schools ($n = 185$) involving random assignment to OST-T2 ($n = 97$) or a treatment-as-usual/waitlist (TAU/WL, $n = 88$) condition. Data were collected between 2017 and 2022; baseline and post-intervention data were collected for about 75% of participants prior to COVID-19 (Nissley-Tsiopinis et al., 2024). The assigned sex of students was 34.2% female. The sample was 12.4% Latiné/Hispanic. As reported by Nissley-Tsiopinis and colleagues (2024), racial composition was 25.9% Black, 3.2% Asian, 0.5% Native Hawaiian/Pacific Islander, 10.3% multi-racial, 56.8% White, and 3.2% not reported or other. An estimated 45.4% of caregivers had a college education or above and 49.7% of participating children likely had a diagnosis of ADHD. Additional information about the sample used in the study of OST-T2 was described in Nissley-Tsiopinis et al. (2024).

The other data set used to examine treatment sensitivity (Aim 2) included students in grades 6–8 participating in a randomized controlled trial investigating the effectiveness of the Homework, Organization, and Planning Skills (HOPS) program(Langberg et al., 2018). All students were diagnosed with ADHD. Students were randomly assigned to HOPS, an alternative intervention (Completing Homework by Improving Efficiency and Focus [CHIEF]), or a TAU/WL condition. Students assigned to the HOPS condition were included in this study because this intervention directly targets OTMP difficulties. Students in CHIEF were not included because this intervention targets a separate, more limited set of behaviors, specifically homework completion and accuracy, and not the broader range of OTMP skills. Students in TAU/WL were included to provide a comparison to HOPS; the sample size for both the HOPS and TAU/WL conditions was 163. Data for this study were collected between 2013 and 2016. The assigned sex of students was 28.8% female. As reported by Langberg and colleagues (2018) , racial composition was 26.4% Black, 0.6% Asian, 1.2% American Indian, 13.5% multi-race, 55.2% White, and 3.1% not reported. An estimated 49% of families had at least one caregiver with some college education or above. Additional information about the sample in the study of HOPS was described in Langberg et al. (2018).

Measure

COSS Data for the COSS were obtained using the originally developed versions of this measure, which included 66 items on the COSS-P and 42 items on the COSS-T (Abikoff & Gallagher, 2009). Because the focus of this study was on developing and validating brief, informant-report measures of OTMP skills, the child self-report version was not used in this study. The parent and teacher versions were administered to children between the ages of 7 and 13 (grades 2–8). Except for the interference items, COSS items are rated on a four-point Likert scale, indicating the degree to which the child is demonstrating an OTMP skill/deficit (1 = *hardly ever or never*; 4 = *just about all the time*). The COSS interference (impairment) items (8 on COSS-P, 4 on COSS-T) assess the extent to which OTMP challenges are interfering with student performance, also rated on a four-point scale (1 = *not at all*; 4 = *very much*). Factor analytic studies of the skills/deficits items have revealed that a bi-factor model with a general factor and three subfactors, consisting of 26 items on the COSS-P and 28 items on the COSS-T, affords adequate fit for both versions (Molitor et al., 2017; Ryan et al., 2024). There is strong support for the internal validity of the general factor and adequate evidence of internal validity for the Task Planning subfactor but not the Organized Actions and Materials Management subfactors. All items of the COSS-P and COSS-T demonstrated pattern coefficients ≥ .50 on the general or Task Planning factor (Ryan et al., 2024). The 26 items of the COSS-P had comparable items on the COSS-T. The 28-item COSS-T included two additional items (*"keeps a neat desk"* and *"completes homework but cannot find homework"*) that were excluded from analyses, because the intent was to identify brief measures that had similar items across informants to facilitate cross-informant comparison.

Data Analyses

Each of the samples was randomly divided into two independent subsamples using SPSS, the first half of which was used for scale development (Study 1) and exploratory factor analyses (Study 2) and the second half of which was used for validation and confirmatory factor analyses (Study 2). Subsamples were equivalent for assigned sex and grade level.

Item Discriminability

An index of discrimination (Ebel, 1979) was computed to evaluate the ability of each skill/deficit item of the 26-item COSS-P and 28-item COSS-T, as well as the four interference items from each scale, to differentiate participants who clearly have OTMP difficulties from those who clearly do not in the general/clinical samples used to examine Aim 1. This index reflects the difference in the proportion of participants who manifest an OTMP skill/deficit/degree of interference (rating of three or four on a scale ranging from one to four) between the subgroup falling in the highest 25% on the total score for the general/non-clinical sample (subgroup high in OTMP deficits) and those scoring in the lowest 25% of this sample (subgroup low in OTMP deficits). Indices ≥ .40 are considered to have good discrimination; those in the .30−.39 range are marginal; and those below .30 are poor (Ebel, 1979).

Item Treatment Sensitivity

Treatment sensitivity for each item (Aim 2) was determined by computing an effect size (Cohen's *d)* for each item using data sets derived from the OST-T2 and HOPS studies. Cohen's *d* was calculated for each item by computing the difference in change scores from baseline to post-intervention for participants in the OST-T2 (or HOPS) condition versus the control condition and dividing this by the pooled standard deviation for the item at baseline. A similar method for computing effect sizes was used in a study of the effectiveness of OST-T2 (Nissley-Tsiopinis et al., 2024). Because actual scores (not model adjusted scores) were used in computing effect sizes, only participants with scores at baseline and post-intervention were included in analyses of treatment sensitivity. For the OST-T2 data sets, sample sizes were 89 and 90 for the caregiver-rated and teacher-rated children, respectively. For the HOPS data sets, sample sizes were 64 and 34 for the caregiver-rated and teacher-rated children, respectively. Because the sample size for the teacher-rated subsample from the HOPS study was small and susceptible to producing unstable estimates, item effect sizes for this subsample were not computed. Effect sizes of moderate magnitude or above (≥ .50) were considered strong; those in the .20 to .49 range were considered fair; and those below .20 were considered weak.

Item Categorization into OTMP Components

To identify the OTMP component associated with each COSS item (Aim 3), a panel of four experts was requested independently to assign items to intervention component. The panel consisted of two clinical psychologists who were involved in the original development and validation of the COSS, one clinical psychologist with extensive experience in conducting research and providing OTMP interventions, and one highly experienced school psychology faculty member who was given an operational definition of each intervention component: Tracking Assignments, Managing Materials, Time Management, and Planning. Three

panel members were able to meet to reconcile differences in item assignments. Each of the panel members contributed as co-authors of this manuscript. Items were classified as aligned with one or more OTMP intervention component, although in two cases items were judged as not aligned with any component. When it was determined that items aligned with more than two intervention components, the item was categorized as "cross-cutting."

Item Selection for Screening and Formative Assessment

Several criteria were applied in identifying skill/deficit items to include in a brief screener. We determined a priori that the screener should include about 10 items to ensure efficiency and provide broad coverage of OTMP components (i.e., inclusion of items for each of the four components and at least one cross-cutting item). Given that items for the FBRM were selected from the pool included in the screener, criteria included a consideration of item treatment sensitivity. The criteria included: (a) there were comparable items across the COSS-P and COSS-T to facilitate cross-informant comparisons; (b) indices of discrimination for the item were ≥.40 for the COSS-P and COSS-T, (c) effect sizes related to treatment sensitivity were ≥.20 in the OST-T2 sample for both parent and teacher ratings and ≥.20 in the HOPS sample for parent ratings, and (d) when possible, at least two items for each OTMP intervention component were retained. In addition, proposed screening items were checked to ensure that pattern coefficients exceeded .50 on the general factor or subfactor 1 (Time Management and Planning) of the COSS-P and COSS-T, based on confirmatory factor analyses of these measures (Ryan et al., 2024). Criteria (a) through (c) were also applied in identifying the interference item to include in the screening form.

Items from the screening version were then selected to be included on the standard FBRM for the purpose of progress monitoring. The investigators determined a priori to include five items in the FBRM, consistent with the approach used by Matta et al. (2020). The primary criterion used to select items for FBRM was item effect size derived from the OST-T2 and HOPS data sets, reflecting the treatment sensitivity of items in response to these child skills training interventions. Indices of discrimination were also considered when selecting these items. In addition, items were intentionally selected to include one item from each OTMP intervention component as well as one cross-cutting item. Finally, a core set of three items, derived from the standard version, was selected to be included in the hybrid FBRM. The two individualized items for the hybrid version were the highest rated items on the 10-item screener that were not included as core items.

Results

Indices of discrimination and item effect sizes for each of the 26 skill/deficit items that were comparable across parent and teacher versions of the COSS for the first half of each sample examined are reported in Table 1. All but four of the 26 items rated by caregivers and teachers in the general/clinical sample had indices of discrimination ≥.40 for both informants, indicating strong discrimination. Three of the four items not having strong discrimination pertained to student forgetfulness. The other item with relatively weak discrimination was the caregiver- and teacher-rated item "*other kids do not like working with child because of disorganization*." All four interference items had indices of discrimination >.40 on both the parent and teacher versions.

Regarding treatment sensitivity, 19 of the 26 skill/deficit items rated by caregivers and teachers had item effect sizes ≥.20 for both informants in the OST-T2 sample and for caregivers in the HOPS sample. Five of these 20 items had item effect sizes ≥.50 for both informants in the OST-T2 sample and for parents in the HOPS sample. The items with relatively high treatment sensitivity (≥.50) values on each indicator included: "*is organized*;" "*puts things in order*;" "*writes down steps before beginning a project"*; "*uses a calendar to know when assignments are due*;" and "*has trouble finding school supplies when needed*." Three of the four interference items had item effect sizes ≥.20 on all indicators; these included "*problems with managing materials," "failure to organize actions/use tools to organize materials," and "problems with planning tasks and assignments."*

The expert panel categorized skill/deficit items into OTMP components in the following way: three items were assigned to Tracking Assignments; nine items to Managing Materials; two items to Time Management; five items to Planning; two items were assigned to both Time Management and Planning, and three items were assigned to three or more OTMP components (referred to as cross-cutting items). In addition, two of the items were determined not to be applicable to any of the OTMP components. One interference item was assigned to each of the following components: Managing Materials management, Tracking Assignments, Planning, and cross-cutting. (See Table 1 for item assignment to OTMP intervention component.)

Based on criteria for screening item selection (in particular, strong indices of discrimination based on caregiver and teacher report; fair to strong indices of treatment sensitivity), 10 skill/deficit items were selected to be included in the screening form. Each of these 10 items met all criteria for item selection. In addition, each of these items had pattern coefficients on the general factor or subfactor 1 (Time Management and Planning) of the COSS-P and COSS-T that exceeded .50 (Ryan et al., 2024). The proposed screener

Table 1 Item Discrimination Indices and Treatment Sensitivity (Item Effect Sizes) for each COSS-P Item and Corresponding COSS-T Item

COSS Skill/Deficit Items	Index of Discrimination General/Clinical Sample Half #1 (Grades 3–8) *N* = 513		Treatment Sensitivity OST-T2 Sample Half #1(Grades 3–5) *N* = 92		Treatment Sensitivity HOPS Sample Half #1 (Grades 6–8) *N* = 82	OTMP Category
	Parent	Teacher	Parent	Teacher	Parent	
Has trouble knowing how to start (P2; T2)	.73	.64	−.54	−.45	−.37	P
Is organized (R-P3; R-T3)*	.91	.93	−.90	−.96	−.94	CC
Runs out of time for schoolwork (P4; T4)	.69	.74	−.32	−.46	−.41	TM
Does not know where to begin a big project (P6; T5)	.83	.58	−.35	−.22	−.34	P
Puts things in order (R-P9; R-T7)**	.79	.91	−1.01	−.92	−.84	MM
Makes lists, schedules, reminders (R-P10; R-T8)	.60	.79	−.85	−1.94	−.26	TA
Does not understand directions for assignments (P13; T11)	.53	.44	−.72	−.41	−.50	P
Hard to find papers may need (P21; T17)	.73	.62	−.42	−.97	−.86	MM
Loses things at school (P26; T22)	.53	.42	−.64	−.62	−.33	MM
Writes down steps before beginning project (R-P29; R-T19)**	.50	.69	−.78	−1.26	−1.42	P
Uses a calendar to know when assignments due (R-P34; R-T25)**	.66	.76	−1.20	−.93	−.69	TA
Uses separate folder for each subject (R-P40; R-T29)	.55	.61	−.41	−.98	−.66	MM
Gets stressed trying to get work done on time (P44; T31)*	.57	.40	−.57	−.33	−.82	TM
Trouble finding school supplies when needed (P53; T9)	.57	.53	−.77	−.54	−.74	MM
Gets stressed trying to keep work organized (P8; T6)	.71	.50	−.23			
Forgets tests (P15; T12)	.27	.25	−.34	−.43	−.53	TA
Checks schoolwork carefully when finished (R-P16; R-T13)	.66	.91	−.17	−.51	−.41	TM, P
Homework assignments are neat (R-P17; R-T14)	.88	.87	−.30	−.58	−.32	CC
Finishes one subject before starting another (R-P18; R-T15)	.62	.81	−.35	−.88	−.15	TM, P
Forgets to bring items to school (P20; T16)	.56	.43	−.26	−.49	−.39	MM
Forgets to return items borrowed (P23; T18)	.31	.25	−.83	−.16	−.36	MM
Does things without thinking of consequences (P24; T20)	.68	.58	−.44	−.08	−.62	NA
Says things without thinking of consequences (P28; T23)	.61	.57	−.39	−.07	−.19	NA
Forgets to return items to school or library (P30; (T21)	.35	.25	−.36	−.35	−.38	MM
Before writing, makes a draft or outline (R-P36; R-T26)	.57	.76	−.44	−.40	.02	P
Other kids do not like to work with child due to disorganization (P50; T33)	.19	.30	−.21	−.36	−.54	CC
COSS Interference Items						
Problems with managing materials (P59; T39)	.77	.77	−.44	−.64	−.99	MM
Problems with remembering schedules and due dates (P60; T40)	.61	.65	−.15	−.03	−.05	TA
Failure to organize actions/use tools to organize materials (P61; T41)	.76	.75	−.85	−.76	−.50	CC
Problems with planning tasks and assignments (P62; T42)	.72	.80	−.56	−.42	−.20	P

R = reverse scored. P for item = Parent-rated. T for item = Teacher-rated. OTMP components: MM = Materials Management; TA = Tracking Assignments; TM = Time Management; P = Planning; CC = Cross-cutting; NA = not applicable to any category. Index of Discrimination ranges: ≥ .4 = strong; .3—.39 = moderate; < .3 = weak. Treatment Sensitivity (item effect size) ranges: ≥ .5 = strong, .2-.49 = fair, < .2 = weak. Items indicated in **bold print** were selected to be included in the proposed screening measure

*indicates items included on the five-item standard formative assessment scale, but not the hybrid formative assessment scale

**indicates items included on both the five-item standard formative assessment scale and the hybrid formative assessment scale

included two items related to Tracking Assignments, three items related to Managing Materials, two items related to Time Management, two items related to Planning, and one cross-cutting item (is organized). In addition, one item (failure to organize actions and use special tools to organize materials interfere with student performance) was selected to screen for OTMP interference with performance. Items selected for the screening measure are indicated in bold print in Table 1.

The five items for the standard FBRM included one item corresponding to each intervention component and a cross-cutting item (is organized). The three core items included in the hybrid FBRM were selected from the standard FBRM. Table 1 identifies items included in the standard and hybrid versions of the FBRM.

Study 2: Measure Validation

Method

Three Aims guided Study 2. Aim 1 focused on examining the factor structure and internal consistency of the COSS screening form developed in Study 1, with a particular focus on determining whether there was justification for scoring a general factor (total score). Aim 2 examined the predictive and convergent validity of the COSS screening form total score in differentiating students with elevated OTMP deficits from those without these deficits as well as its associations with measures of school and executive functioning. Aim 3 examined the treatment sensitivity of the screening form and a range of options for monitoring student progress using FBRMs.

Participants

Participants included in analyses for Study 2 were derived from the three samples described earlier for Study 1. To address Aim 1 (factor structure, internal consistency) participants in the general and clinical sample were included. For Aim 2 (predictive and convergent validity), participants from the general and clinical sample and intervention studies evaluating OST-T2 and HOPS were included. For Aim 3, only participants from the OST-T2 and HOPS RCT samples were included.

Data Analyses

Aim 1: Factor Structure of COSS Screener: Exploratory Factor Analysis (EFA) Using the first half of the general and clinical COSS-P ($n=513$) and COSS-T ($n=475$) samples, initial EFAs were conducted in MPlus version 8.11 (Muthén & Muthén, 2023) to examine the latent factor structure of the 10 COSS skills/deficit items selected for the screener in study one. Preliminary analyses to determine the appropriateness of factor analyses of the data, including Bartlett's Test of Sphericity (Bartlett, 1951) and Kaiser–Meyer–Olkin statistic (KMO; Kaiser, 1974) calculations, were conducted in SPSS version 31 (IBM Corp., 2023). Items were treated as categorical and analyzed using weighted mean and variance adjusted estimation (WLSMV; Beauducel & Herzberg, 2006), which models polychoric correlations among items and accommodates missingness via pairwise-present analyses under missing-at-random assumptions. Item-level missingness was examined descriptively and found to be low (<1%) across all 10 proposed screening items for both parent- and teacher-rated forms. Models used an oblique rotation (GEOMIN estimation method; Muthén & Muthén, 2017), allowing correlated factors. One-, two-, and three-factor solutions were examined, with several procedures informing the number of factors to extract (Kline, 2016). First, global model fit statistics, including Comparative Fit Index (CFI), Tucker-Lewis Index (TLI; CFI and TLI≥.95 indicates good fit; Whitaker & Schumacker, 2022), root mean square error of approximation (RMSEA;≤.08 indicates good fit; Browne & Cudeck, 1993), and standardized root mean square residual (SRMR;≤.05 indicates good fit; Hu & Bentler, 1999), were examined. Second, scree plots and an examination of eigenvalues were used (>1.0 for factor retention, Kaiser, 1960). Third, parallel analysis was conducted to account for sampling error associated with eigenvalue-greater-than-1 approach (Horn, 1965). Finally, criteria for factor retention included a minimum of three items with patterns coefficients greater than 0.40 and McDonald's Omega>0.70 (McDonald, 1999).

Aim 1: Factor Structure of COSS Screener: Confirmatory Factor Analysis (CFA) CFAs were conducted with the second half of the general and clinical samples for COSS-P ($n=516$) and COSS-T ($n=478$) to confirm the factor structures of the 10 COSS screener skill/deficit items identified via EFAs. WLSMV estimation in MPlus (Beauducel & Herzberg, 2006) was used to conduct CFAs. Moreover, consistent with our aim to identify a general factor on the COSS screening form with potential clinical utility and prior evidence of a general OTMP factor based on factor analytic studies of longer versions of the COSS (e.g., Molitor et al., 2017; Ryan et al., 2024), bi-factor solutions were examined. Model fit was first determined using conventional global fit indices: CFI and TLI≥.95 (Whitaker & Schumacker, 2022); RMSEA≤.08 (Browne & Cudeck, 1993); and SRMR≤.05 (Hu & Bentler, 1999). To address limitations of fixed cutoff values, dynamic fit indices (DFIs; McNeish & Wolf, 2023)

were subsequently computed for each CFA model, generating simulation-based, model-specific thresholds tailored to the sample size, estimator, and ordinal indicators. Omega and omega hierarchical coefficients were calculated to determine the internal reliability of factors in multi-dimensional models (Watkins, 2017). Omega coefficients > 0.70 were considered indicative of acceptable levels of factor reliability for non-hierarchical models. In hierarchical, bi-factor models, omega hierarchical coefficients > 0.50 indicate acceptable levels of reliable variance in the total factor, distinct from that of the subfactors, and omega hierarchical coefficients > 0.75 indicate high levels of reliable variance associated with the total factor (Riese, 2012; Watkins, 2017).

Aim 2: Predictive Validity of COSS Screeners: Receiver Operating Characteristic (ROC) Analyses ROC analyses were also conducted with the second half of the general and clinical samples for the COSS-P ($n = 516$) and COSS-T ($n = 478$) to determine the predictive validity of scores of interpretable factors identified and confirmed via factor analysis of the 10-item COSS screeners. Specifically, ROC curves were plotted to evaluate the ability of interpretable factor scores to discriminate between children who demonstrated evidence of substantial and impairing OTMP deficits (i.e., scored at least one *SD* above the mean total score on the 26-item version of the COSS-P and 28-item version of the COSS-T [Ryan et al., 2024] *and* demonstrated endorsement of at least one item assessing OTMP interference with performance) and those who did not meet these criteria. These analyses were conducted within informant, because the caregiver and teacher samples generally were non-overlapping. Area under the curve (AUC) indices were calculated with values ≥ .90 indicating excellent discrimination, ≥ .80 indicating good discrimination, ≥ .70 indicating fair discrimination, and < .70 indicating poor discrimination (Swets et al., 2000). Sensitivity, specificity, positive predictive power (PPP; i.e., percentage of participants scoring above a specified cut-point who met criteria for substantial, impairing OTMP), negative predictive power (NPP; i.e., percentage of individuals scoring below the cut-point who did not meet criteria for substantial, impairing OTMP), and total predictive power (TPP; i.e., combined PPP and NPP) were examined to determine optimal cut-points. Additionally, a kappa correction was applied to TPP (cTPP) to adjust for low base rates (Youngstrom, 2014). Optimal cut-points for screening were considered as those with sensitivity and specificity values of at least 0.80 and a cTPP value of at least .60 (Koshy et al., 2016).

Aim 2: Convergent Validity of COSS Screeners: Correlations with Measures of School and Executive Functioning Convergent validity of the parent and teacher versions of the 10-item COSS screener (COSS10) was evaluated with the second half of the OST-T2 ($n = 91$; Nissley-Tsiopinis et al., 2024) and HOPS ($n = 82$; Langberg et al., 2018) RCT samples. Specifically, among OST-T2 participants (i.e., grades 3–5), bivariate correlations of the parent and teacher COSS10 total scores with scores on the following measures were examined: parent-completed Homework Problems Checklist (HPC; Anesko et al., 1987); teacher-completed Homework Performance Questionnaire (HPQ; Power et al., 2007), teacher-completed Academic Competence Evaluation Scales (ACES; DiPerna & Elliot, 2000); and teacher-completed Academic Proficiency Scale (APS; Abikoff et al., 2013). Among HOPS participants (i.e., grades 6–8), bivariate correlations between the parent and teacher COSS10 total scores and the following measures were examined: parent-completed HPC; parent- and teacher-completed HPQ (HPQ-P, HPQ-T); and parent-completed Behavioral Regulation Index (BRI) and Metacognition Index (MI) from the Behavior Rating Inventory of Executive Function (BRIEF; Gioia et al., 2000). For descriptions of measures and psychometric properties, see Supplemental Online Material.

Aim 3: Treatment Sensitivity of COSS Screeners and FBRMs Using pre- and post-intervention scores for the second half of the OST-T2 ($n = 91$; Nissley-Tsiopinis et al., 2024) and HOPS ($n = 82$; Langberg et al., 2018) RCT samples, treatment sensitivity of the 10- and 11-item (COSS11, with interference item) versions of the COSS screener (identified in Study 1 and confirmed via CFA in Study 2) was examined. Item scores were summed to calculate total scores for the COSS10 (range: 10–40) and COSS11 (range: 11–44). In addition, treatment sensitivity was determined for three variations of 5-item FBRMs developed for progress monitoring: (1) a standard measure; (2) a hybrid measure with 3 core items and 2 individualized items; and (3) an individualized measure. The five items selected for the standard FBRM and the three items selected to be included in the hybrid version are indicated in Table 1. The two individualized items for the hybrid measure were selected based on item ratings indicating the most severe skill deficits at baseline (items were reverse scored, as indicated). Consistent with previous research on individualized formative behavior assessment measures (Volpe et al., 2009), when more than two items received maximum ratings (i.e., "*just*

about all of the time," item score = 4), items were randomly selected among the pool with ratings of 4. When fewer than two items received ratings of 4, random selection was used among items rated 3. These procedures were used for each participant until two items with the highest baseline rating were selected. Finally, identical procedures for selecting the two individualized items in the hybrid measure were used to select the five items for the fully individualized measure. The five items selected for each version of the proposed FBRMs were summed (scores range: 5–20) to create pre- and post-intervention scores for the OST-T2 and HOPS datasets. Between-group (i.e., OTMP intervention vs. waitlist control) differences in post-intervention scores on the COSS10, COSS11, and each version of the formative assessment measures were evaluated via one-way ANCOVA, controlling baseline scores. Cohen's *d* effect sizes were also calculated to determine the magnitude of the mean differences in change scores between groups.

Results

Aim 1: Factor Structure of COSS Screeners: Findings from Exploratory Factor Analyses

COSS10-P Bartlett's Test of Sphericity yielded a statistically significant result ($p < .001$; Bartlett, 1951), indicating that the items were sufficiently correlated and that factor analysis of the data was justified. Examination of the KMO statistic (.90) further supported the appropriateness of the data for factor analysis. Eigenvalues from the correlation matrix, examination of the scree plot, and parallel analysis suggested up to two factors for extraction. The three-factor solution yielded acceptable model fit across indices (CFI = .995; TLI = .988; RMSEA = .062; SRMR = .023), as did the two-factor (CFI = .987; TLI = .977; RMSEA = .082; SRMR = .036), although the three-factor model was superior. The one-factor (CFI = .947; TLI = .932; RMSEA = .144; SRMR = .090) solutions did not demonstrate adequate model fit across all indices. Further, at least three items loaded onto each factor in both two- and three-factor solutions, each factor met minimum reliability criteria (i.e., ωs > .70; Brown, 2002) and modest inter-factor correlations (.55-.67) emerged. Given their support across criteria, both two- and three-factor solutions were examined in subsequent confirmatory analyses of identical item structures in correlated factor and bi-factor models (see Supplementary Table 1 for results of the one-, two-, three-factor solutions derived from EFAs of the COSS10-P screener items).

COSS10-T Bartlett's Test of Sphericity ($p < .001$) and the KMO statistic (.91) indicated the data were appropriate for factor analyses. Eigenvalues from the correlation matrix, examination of the scree plot, and parallel analysis suggested up to two factors. However, only the three-factor solution yielded acceptable model fit across indices (CFI = .996; TLI = .990; RMSEA = .082; SRMR = .016); the two-factor (CFI = .985; TLI = .974; RMSEA = .133; SRMR = .045) and one-factor solutions (CFI = .966; TLI = .956; RMSEA = .173; SRMR = .086) did not demonstrate adequate model fit across all indices. Additionally, at least three items loaded onto each factor in both two- and three-factor solutions; each factor met minimum reliability criteria (i.e., ωs > .70; Brown, 2002); and moderate to high inter-factor correlations (.63-.81) emerged. As such, similar to the COSS10-P, both two- and three-factor solutions were examined in subsequent confirmatory analyses of identical item structures in correlated factor and bi-factor models (see Supplementary Table 2 for results of the one-, two, three-factor solutions derived from EFAs of the COSS10-T screener items).

Aim 1: Factor Structure of COSS Screeners: Findings from Confirmatory Factor Analyses

COSS10-P A bi-factor model with three and two subfactors and non-hierarchical models with three factors, two factors, and one factor were examined with the second half of the general and clinical sample for COSS-P items ($n = 516$). The bi-factor solution with two subfactors emerged as the best-fitting model based on evaluation of conventional (see Table 2) and dynamic (see Supplementary Table 3) fit indices. Explained common variances were .71 for the general factor, .16 for subfactor 1, and .13 for subfactor 2. The omega hierarchical coefficient was high for the general factor (.81) and low for each subfactor (.33 and .21 for subfactors 1 and 2, respectively). Items loading onto the first subfactor include those representing OTMP skills associated with materials management and assignment preparation. Items loading onto subfactor 2 include those representing OTMP skills associated with time management and assignment completion. See Table 3 for pattern coefficients for the COSS10-P bi-factor model with two subfactors.

COSS10-T A bi-factor model with three and two subfactors and non-hierarchical models with three factors, two factors, and one factor were examined with the second half of subsample of the general and clinical for COSS-T items ($n = 478$). The bi-factor solution with three and two subfactors yielded acceptable model fit, but the model with three subfactors was superior in evaluations of both conventional (see Table 2) and dynamic (see Supplementary Table 3) fit indices. Explained common variances were .79 for the general factor and .03, .09, and .10 for subfactors 1–3, respectively. The omega hierarchical coefficients were high for the general factor (.90) and low for subfactors 1–3 (.01, .26,

Table 2 Fit Statistics for Confirmatory Factor Analyses of COSS-P and COSS-T

Model	*N*	χ^2	*df*	CFI	TLI	RMSEA [90% CI]
COSS-P						
1. One factor	516	566.34	35	.936	.918	.172 [.159, .184]
2. Two factor	516	317.85	34	.966	.955	.127 [.155, .140]
3. Three factor	516	285.34	32	.969	.957	.124 [.111, .137]
4. Bi-factor with two subfactors	516	90.47	25	.992	.986	.071 [.056, .087]
5. Bi-factor with three subfactors	516	199.71	25	.979	.962	.116 [.102, .132]
COSS-T						
1. One factor	478	545.41	35	.969	.961	.178 [.165, .191]
2. Two factor	478	397.53	34	.979	.972	.150 [.137, .163]
3. Three factor	478	219.99	32	.989	.985	.111 [.097, .125]
4. Bi-factor with two subfactors	478	99.27	25	.996	.992	.079 [.063, .095]
5. Bi-factor with three subfactors	478	80.69	25	.997	.994	.068 [.052, .085]

Note. χ^2=Chi-Squared, CFI=Comparative Fit Index, TLI=Tucker-Lewis Index, RMSEA=Root Mean Square Error of Approximation, CI=Confidence Interval, COSS-P=Children's Organizational Skills Scale, Parent Version, COSS-T=Children's Organizational Skills Scale, Teacher Version. All χ^2 tests were statistically significant, $p<.001$

Table 3 Pattern Coefficients for the COSS-P Bi-Factor Model Derived from Confirmatory Factor Analyses

	General OTMP	Materials Management/ Assignment Preparation	Time Management/ Assignment Completion
2. Has trouble knowing how to start	.769		.367
3. Is organized	.843	.285	
4. Runs out of time for schoolwork	.604		.678
9. Puts things in order	.774	.425	
10. Makes lists, schedules, reminders	.589	.679	
21. Hard to find papers may need	.884		.142
29. Writes down steps before beginning project	.595	.480	
34. Uses a calendar to know when assignments due	.464	.438	
44. Gets stressed trying to get work done on time	.434		.535
53. Trouble finding school supplies when needed	.882		.111
Explained Common Variance	.71	.16	.13
ω	.94	.90	.92
ω_h	.81	.33	.21

COSS-P=Children's Organizational Skills Scale, Parent Version. Negative pattern coefficients indicate that items need to be reverse scored. ω refers to the omega coefficient or reliability. ω_h refers to the omega hierarchical coefficient, which is used in hierarchical factor analytic models, including bi-factor models

and .35, respectively). Items loading onto subfactors 1–3 represent OTMP skills associated with materials management, assignment tracking, and time management and planning, respectively. See Table 4 for pattern coefficients for the COSS10-T bi-factor model with three subfactors.

Aim 2: Predictive Validity of COSS Screeners: Findings of ROC Analyses

COSS10-P The AUC for the COSS10-P was significantly different from the null (.50), demonstrating excellent differentiation of youth who met criteria for substantial and impairing OTMP difficulty and those who did not (AUC=.947, $p<.001$ [95% CI: .925–.969]). Clinical utility statistics aiding in the identification of an optimal cut-point score on the COSS10-P are presented in Table 5. The optimal cut-point for the COSS10-P was 28.5, which was associated with strong sensitivity (.834), specificity (.938), and TPP (cTPP=.781).

COSS10-T The AUC for the COSS10-T was significantly different from the null (.50) and indicated excellent discrimination between participants who met criteria for substantial and impairing OTMP difficulty and those who did not (AUC=.982, $p<.001$ [95% CI .973–.992]). Clinical utility statistics (see Table 5) indicated the optimal cut-point on the COSS10-T was 27.5, which was associated with strong sensitivity (.921), specificity (.948), and TPP (cTPP=.854).

Aim 2: Convergent Validity of COSS Screeners: Bivariate Correlations with School and Executive Functioning Measures

COSS10-P. Strong, positive bivariate associations emerged between the COSS10-P the parent-completed HPC (higher scores indicate more homework difficulties) in both the OST-T2 ($r=.68$) and HOPS ($r=.61$) RCT samples ($ps<.001$). COSS10-P total scores were also significantly correlated with scores on the HPQ-T (higher scores indicate better homework performance; $r=-.23$, $p=.033$) among OST-T2 participants but not HOPS participants ($r=-.10$, $p=.386$). Among OST-T2 participants, teacher-rated academic

Table 4 Pattern Coefficients for the COSS-T Bi-factor Model Derived from Confirmatory Factor Analyses

COSS-T Items	General OTMP	Materials Management	Assignment Tracking	Time Management/ Planning
2. Has trouble knowing how to start	.750			.377
3. Is organized	.953	.135		
4. Runs out of time for schoolwork	.699			.555
7. Puts things in order	.933	.327		
8. Makes lists, schedules, reminders	.803		.352	
9. Trouble finding school supplies when needed	.892	.125		
17. Hard to find papers may need	.910	.248		
19. Writes down steps before beginning project	.722		.429	
25. Uses a calendar to know when assignments due	.680		.617	
31. Gets stressed trying to get work done on time	.343			.569
Explained Common Variance	.79	.03	.09	.10
ω	.96	.92	.90	.84
ω_h	.90	.01	.26	.35

COSS-T = Children's Organizational Skills Scale, Teacher Version. Negative pattern coefficients indicate that items need to be reverse scored. ω refers to the omega coefficient of reliability. ω_h refers to the omega hierarchical coefficient, which is used in hierarchical factor analytic models, including bi-factor models

functioning was significantly associated with COSS10-P total scores when examining the ACES ($r = -.24$, $p = .032$) but not the APS ($r = -.17$, $p = .119$). Among HOPS participants, total scores on the COSS10-P were significantly correlated with HPQ-P ($r = -.53$, $p < .001$) and BRIEF-MI (higher scores indicate worse executive functioning abilities; $r = .61$, $p < .001$). See Supplementary Tables 4 and 5 for correlation matrices for OST-T2 and HOPS participants, respectively.

Table 5 Clinical Utility Estimates for the Second Half of the Combined Clinical and Non-Clinical Sample on High OTMP Difficulty Status

Cutoff	PPP	NPP	cTPP	Sensitivity	Specificity
COSS10-P					
24.5	0.616	0.957	0.563	0.941	0.693
25.5	0.675	0.950	0.635	0.923	0.768
26.5	0.723	0.939	0.680	0.899	0.820
27.5	0.801	0.935	0.749	0.882	0.885
28.5	0.876	0.915	0.781	0.834	0.938
29.5	0.954	0.885	0.776	0.757	0.981
30.5	0.957	0.839	0.678	0.639	0.985
31.5	0.990	0.809	0.612	0.550	0.997
32.5	1.000	0.773	0.506	0.438	1.000
COSS10-T					
24.5	0.639	1.000	0.667	1.000	0.782
25.5	0.696	0.993	0.726	0.984	0.834
26.5	0.785	0.980	0.799	0.952	0.899
27.5	0.873	0.969	0.854	0.921	0.948
28.5	0.914	0.943	0.836	0.849	0.969
29.5	0.950	0.904	0.770	0.730	0.985
30.5	0.976	0.878	0.712	0.643	0.994
31.5	0.974	0.859	0.654	0.579	0.994
32.5	0.969	0.835	0.572	0.492	0.994

High OTMP Difficulty = children who scored at least one *SD* above the non-clinical sample mean total score on the 26- (parent) and 28-item (teacher) COSS (Ryan et al., 2024). PPP = positive predictive power. NPP = negative predictive power. cTPP = kappa correction for total predictive power.

COSS10-T Total scores on the COSS10-T were only significantly associated with teacher-completed measures among OST-T2 participants (each in the expected direction): HPQ-T ($r = -.29$, $p = .006$); APS ($r = -.33$, $p = .002$); and ACES ($r = -.33$, $p = .001$). Alternatively, among HOPS participants, total scores on the COSS10-T were significantly associated with both parent and teacher ratings of homework performance: HPC ($r = .26$, $p = .048$); HPQ-T ($r = -.64$, $p < .001$); and HPQ-P ($r = -.45$, $p < .001$). See Supplementary Tables 4 and 5 for correlation matrices for OST-T2 and HOPS participants, respectively.

Aim 3: Treatment Sensitivity of COSS Screeners and FBRMs

COSS-P. Significant main effects of treatment condition emerged for post-treatment total scores on each parent-completed screener (i.e., COSS10, COSS11) and FBRM (i.e., standard, hybrid, individualized) for both OST-T2 and HOPS samples, after controlling for baseline functioning (*p*s < .05; see Table 6). All effects were in the large range (*d*s range: −1.20 to −1.63) and favored youth assigned to the OTMP intervention groups. Among elementary students who

received OST-T2, pre-to-post-treatment effects appeared to be higher among fully individualized (d=−1.63) and hybrid (d=−1.50) parent-completed FBRMs compared to the standard FBRM (d=−1.20). Alternatively, among middle school students who received HOPS, the parent-completed standard measure appeared to have higher treatment sensitivity (d=−1.57) relative to the hybrid (d=−1.23) and fully individualized (d=−1.29) measures.

COSS-T Significant main effects were observed on total scores of the teacher-completed screeners (i.e., COSS10, COSS11) and FBRMs (i.e., standard, hybrid, individualized) at post-treatment among OST-T2 participants, after controlling for baseline functioning (ps<.001; see Table 6). All effects were in the large range (ds range: −1.30 to −1.75) and favored youth assigned to the OST-T2 intervention. Consistent with findings for the parent-completed measures in the OST-T2 sample, pre-to-post-treatment effects appeared to be larger among fully individualized (d=−1.75) and hybrid (d=−1.58) FBRMs compared to the standard measure (d=−1.37).

Table 6 Summary of One-Way Analyses of Covariance Effects and Effect Sizes for Contrasts Between OTMP Interventions and TAU/WL in Study 2

Dependent Variable	F (df)	p	Partial η2	d
OST-T2 Teacher				
COSS10 screener	$F(1, 80)=26.33$	<.001	.248	−1.34
COSS11 screener	$F(1, 80)=25.07$	<.001	.239	−1.30
Standard FBRM	$F(1, 80)=23.29$	<.001	.226	−1.37
Hybrid FBRM	$F(1, 80)=19.44$	<.001	.196	−1.58
Individualized FBRM	$F(1, 80)=19.88$	<.001	.199	−1.75
OST-T2 Parent				
COSS10 screener	$F(1, 82)=31.07$	<.001	.275	−1.28
COSS11 screener	$F(1, 82)=32.17$	<.001	.282	−1.26
Standard FBRM	$F(1, 82)=29.96$	<.001	.268	−1.20
Hybrid FBRM	$F(1, 82)=43.89$	<.001	.349	−1.50
Individualized FBRM	$F(1, 82)=41.54$	<.001	.344	−1.63
HOPS Parent				
COSS10 screener	$F(1, 61)=18.46$	<.001	.232	−1.27
COSS11 screener	$F(1, 50)=16.41$	<.001	.247	−1.31
Standard FBRM	$F(1, 56)=22.55$	<.001	.287	−1.57
Hybrid FBRM	$F(1, 61)=12.40$	<.001	.169	−1.23
Individualized FBRM	$F(1, 61)=9.39$	.003	.133	−1.29

This table indicates differences between the OTMP intervention and treatment-as-usual/waitlist (TAU/WL) conditions at post-intervention, controlling for baseline scores, for each of the screening versions and formative behavior rating measures (FBRM) examined in this study. COSS refers to the Children's Organizational Skills Scale. OST-T2 is the Organizational Skills Training – Tier 2 intervention. HOPS is the Homework, Organization, and Planning Skills program. Values>.14 for partial η2 reflect large effect sizes. Values greater than 0.80 for Cohen's d reflect large effects

Discussion

OTMP skills are behavior manifestations of executive functioning that have been shown to be associated with homework and academic performance in the future (Langberg et al., 2013). Improvements in OTMP skills are leading indicators of academic achievement and may be associated with decreased family conflict and better attitudes toward school, making them important targets for intervention (Abikoff et al., 2013; Evans et al., 2021; Pfiffner et al., 2013; Power et al., in press). Highlighting the critical importance of OTMP skills, numerous studies have demonstrated the effectiveness of interventions targeting OTMP skills in improving homework and academic performance at the elementary, middle school, and high school levels (Abikoff et al., 2013; DuPaul et al., 2021; Evans et al., 2016; Nissley-Tsiopinis et al., 2024; Sibley et al., 2016).

The crucial link between OTMP skills and academic performance demonstrates the need for well-developed tools to assess OTMP skills. The COSS parent and teacher versions have been highly useful tools for determining eligibility for OTMP interventions and evaluating outcomes. However, these measures are relatively long and may not be practical for use in screening children for intervention eligibility and monitoring student progress in response to intervention. The purpose of this study was to develop brief measures of OTMP that are efficient and feasible to use in screening and progress monitoring for students in grades 3 through 8.

In this study our team was successful in developing a 10-item (and 11-item) screening measure consisting of items that were strong in discriminating children with high and low levels of OTMP skill and detecting improvement in OTMP skills in response to targeted intervention. The items selected for the brief measures in this study demonstrated utility with students in upper elementary school (grades 3–5; study of OST-T2) and middle school (grades 6–8; study of HOPS). The 10-item screening measure included 9 items assessing the four components of OTMP intervention (keeping track of assignments, managing materials, managing time, planning) and one cross-cutting item (is organized). The 11-item version also included 1 item assessing interference with performance. The 10 items assessing OTMP behaviors showed strong evidence of reliability, structural validity and clinical utility. A confirmatory factor analysis of the COSS-P and COSS-T demonstrated good fit to a bi-factor model with a robust general factor having excellent internal consistency, supporting interpretation of a total score on each measure. ROC analyses indicated that each measure was excellent at differentiating children with elevated and impairing OTMP deficits from those who did not. In addition, the screening measures demonstrated convergent validity with measures of school and executive functioning as well as excellent treatment sensitivity; effect sizes for the COSS-P and COSS-T exceeded −1.20 when

examining change in response to OTMP interventions, including OST-T2 for students in grades 3–5 and HOPS for students in grades 6–8.

Based on research using multi-item scales for formative assessment (Volpe & Briesch, 2015; Volpe et al., 2009, 2023) and research in this study on COSS screening measures, our team derived three FBRMs for use in progress monitoring: a standard 5-item measure, an individualized 5-item measure, and a hybrid measure including three core items and two individualized items derived from the screeners. Each method was shown to have a high degree of treatment sensitivity when examined in the context of randomized controlled studies of OST-T2 and HOPS with effect sizes reflecting change from baseline to post-intervention at or exceeding effect sizes of −1.20.

Clinical Implications

This study has numerous implications for practice. First, the study supports the use of the 10-item COSS-P and COSS-T as a screening tool. The general factor of the COSS-P and COSS-T demonstrated strong factorial integrity and reliability, justifying the scoring and interpretation of a total score. Based on clinical utility statistics, the optimal cut-point for screening was 28.5 for the COSS10-P and 27.5 for the COSS10-T. These cut-points achieved optimal balance in sensitivity and specificity (as well as predictive power). If the goal is to maximize the ability to predict (rule in) cases with elevated and impairing OTMP deficits, a slightly higher score (29.5 for COSS10-P; 29.5 for COSS10-T) is recommended, and if the purpose is to maximize the ability to accurately rule out cases without elevated and impairing OTMP deficits, a slightly lower score (27.5 for COSS10-P) and the same score (27.5 for COSS10-T) is indicated. Consistent with diagnostic criteria for diagnosing mental health disorders (American Psychiatric Association, 2022), we advise that determinations of risk for substantial, impairing OTMP challenges be based on two considerations: (a) scores that exceed identified thresholds on the 10-item COSS, and (b) endorsement of the interference (11th) item ("*fails to organize actions and use special tools to organize materials interfere with student performance"*).

Second, although subfactors on the COSS-10-P and COSS10-T were identified in bifactor models, each of these subfactors was weak (low omega hierarchical coefficients), which does not provide justification for the scoring and interpretation of the subfactors. However, in clinical practice consideration of scores on the subfactors, as well as individual items, may be useful for intervention planning such as potential identifying targets for OTMP skills training.

Third, this study was successful in developing three formative behavioral rating measures (FBRMs): (1) a standard five-item method using the same items for all students, (2) an individualized five-item method using items tailored for each student that reflect the OTMP behaviors of greatest concern, and (3) a hybrid method including three core items used with all students and two items tailored to each student. Each of these FBRMs demonstrated excellent treatment sensitivity when applied using the COSS10-P and COSS10-T. Based on findings of this study using data from the OST-T2 sample as well as prior research (Volpe & Briesch, 2015), the individualized approach may be the most sensitive to intervention, although findings using data from the HOPS study did not corroborate this pattern. The individualized method may be most useful and feasible to use when monitoring progress for one or a small group of students. The standard method likely is easier for teachers to use in a classroom setting because it does not require the additional step of identifying individual items for each student. The standard method also enables educators to make comparisons across students. The hybrid approach leverages assets of the standard and individualized approaches.

Fourth, as indicated in Table 6, both the parent and teacher versions of the screeners demonstrated high levels of treatment sensitivity. In addition to identifying students at risk for substantial, impairing OTMP challenges, the screeners may have utility as outcome measures administered at baseline, post-intervention, and follow-up. Because these measures are brief and psychometrically sound, they yield accurate estimates of OTMP functioning in a manner that minimizes the burden on caregivers and teachers in completing measures.

Study Limitations

Several limitations are important to note. First, the evaluation of treatment sensitivity was based solely on differences between conditions in ratings at baseline compared to post-intervention. A stronger test of treatment sensitivity would be to examine rates of change based on assessments at multiple points in time. Second, given that analyses of the psychometric soundness of FBRMs included only two data points (baseline, post-intervention), it was not possible to conduct generalizability and dependability studies that provide more useful estimates of the soundness of these metrics (Briesch et al., 2014). Third, response options for COSS items were on a four-point scale, which is acceptable for screening and outcome assessment, but may limit treatment sensitivity when used in formative assessment. In future research and practice, it is recommended that response options for FBRM measures use a seven-point scale (Matta et al., 2020; Volpe et al., 2023). Fourth, items

for the five-item individualized FBRM and the two individualized items of the hybrid FBRM were generally identified by randomly selecting among items with the highest ratings (reverse scoring as indicated). We used this method because we were working with existing data sets. In practice, it is recommended that individualized items included in standard and hybrid FBRMs be selected through collaboration and shared decision making among clinicians, caregivers, and teachers. Fifth, characteristics of the samples used in this study place limits on the generalizability of findings. The large general and clinical sample used in analyses of item discriminability and the factor structure of COSS screening measures was collected between 2004 and 2007 when the US population was less diverse than it currently is. Although children included in the general and clinical samples generally reflected characteristics of the US population in 2000, the caregivers of non-English speaking children were not included, and caregivers of Black children were underrepresented. For the OST-T2 and HOPS samples, children who were Black and White were overrepresented and those who were Asian, American Indian, and Hispanic/Latino were underrepresented. In addition, the samples included in the OST-T2 and HOPS studies were drawn from one metropolitan area (different for each study) on the East coast. All study samples underrepresented children living in poverty or deep poverty. Sixth, the a priori decision to limit the screening measure to 10 items excluded a few items that had acceptable levels of discriminability and treatment sensitivity. Feasibility considerations strongly influenced this decision as inclusion of additional items may have resulted in a more burdensome measure for caregivers and teachers. Seventh, the research described in this paper was conducted between 2004 and 2022, and almost all of the data were collected prior to the onset of the COVID-19 pandemic. The findings and clinical implications of this study should be considered in light of this limitation. Some items included in the current versions of the COSS may not account for digital technologies introduced recently to support OTMP skill use, although many of the items on the COSS assess behaviors that are not specific to paper-based vs. technology-supported learning and appear to have face validity in both environments (e.g., "runs out of time for schoolwork," "has trouble knowing how to start," and "puts things in order"). In a current replication study of the HOPS program, checklists have been modified for the intervention that included OTMP behaviors related to the use of online/digital applications. These checklists may be useful for informing modifications to the COSS item pool. It is recommended that researchers engage in partnerships with caregivers, teachers, and students to determine if any adjustments in item wording are needed and evaluate the validity of modified versions. Finally, outcomes for students in the combined general/clinical, OST-T2, and HOPS samples were assessed at various times during the school year, which may have had some impact, likely minor, on estimates of clinical utility for screening and treatment sensitivity. In future studies using larger samples, it is recommended that researchers examine the impact of time of school year and level of schooling (i.e., elementary vs. middle school), on optimal cut-points for screening and treatment sensitivity.

Conclusions

A growing body of research is affirming the strong association between OTMP skills and academic performance. Numerous studies have demonstrated the effectiveness of OTMP skills training interventions for children and adolescents, in particular those with ADHD. There is an increasing need for assessment tools to identify children who may benefit from these interventions and monitor student progress in response to intervention. This study focused on the development of efficient tools for screening OTMP skills deficits and monitoring response to OTMP skills training programs. The screening tools included 11-item caregiver-report and teacher-report versions, including one item assessing OTMP behaviors interfering with performance. Each of the screeners yields a total score that is reliable, highly effective in differentiating children with and without OTMP deficits, and highly sensitive to the effects of OTMP training interventions at the elementary and middle school levels. The study was also successful in developing 5-item formative assessment measures, including a standard version that can be used with all children, an individualized version with items tailored to each student, and a hybrid version including 3 core items and 2 individualized items. Each version was shown to be highly sensitive to the effects of OTMP training programs. Research with diverse populations of students examining the generalizability and dependability of formative measures is strongly encouraged.

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